

Chapter 3

Observed Regularities

Beyond the classification, several quantitative regularities were noted across all trials.

3.0.1 Internal Color Periodicity

Along any directional line, the color pattern of heights repeats periodically without variation. For instance, a line with $(2, 1)$ always follows the sequence (brown, white, brown) from left to right.

3.0.2 Frequency of Occurrence

Lines whose slope is close to horizontal or vertical appear less frequently across trials. The vast majority of observed lines are horizontal, vertical, or at a 45° angle (slope ± 1). Lines with slopes $1/2$, $1/3$, 2 , or 3 are comparatively rare.

3.0.3 Vector Sum at Intersections

When multiple directional lines meet at a single intersection point, the sum of their unit direction vectors equals zero. This suggests a form of local conservation governing the pattern geometry.